

# Practice Exam 1

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## True/False/Uncertain, Explain

For Questions 1-5, answer true/false/uncertain and explain. Make sure you provide an explanation even if the statement is true, as most of the points are awarded for the explanation!

1. If a change in government policy results in greater production of both consumer goods and investment goods, the economy was not employing all of its resources before the policy change.
2. A budget constraint's slope is a function of the prices of the two goods.
3. If we observe the price in a given market increase, then the demand for that good rose.
4. High-income people will sometimes pay higher prices at convenience stores for goods that are available at discount stores. One reason they may do this is because crowded and understaffed discount stores impose higher time costs relative to convenience stores.
5. Specialization and trade allow individuals to shift their PPFs outward.

## Problems

1. You have \$50 of income to spend on two goods, candy and decorations. Suppose that the price of candy,  $P_C$ , is \$1 and the price of decorations is  $P_D$ , is \$2.
  - (a) Draw your budget constraint.
  - (b) What is the opportunity cost of candy? Of decorations?
  - (c) Suppose that the price of candy increases to \$1.25. Draw the budget constraint. Does your opportunity cost change? Does your consumption possibilities change? Why or why not?
  - (d) Suppose instead that your income falls to \$40 (so,  $P_C$  returns to \$1). Draw the new budget constraint. Compared to parts a and b, do your opportunity costs change? Does your possible consumption bundles change? Why or why not?
2. Consider the market for face masks. With the onset of COVID-19, mask wearing has been encouraged, and in some places mandated. At the same time, many individuals have lost their jobs or are more uncertain about the stability of their job.
  - (a) On one clearly labeled graph for the market for face masks, show the effects on equilibrium price and quantity. If either price or quantity are indeterminate, explain what you would expect to happen to that variable and why.
  - (b) Explain (in words) the changes in your graph. Make sure you:
    - explicitly name the shifter(s) you refer to
    - discuss how the market transitions from one equilibrium to the next.
3. Two friends, Monica and Chandler, enjoy baking bread and making apple pies. Monica takes two hours to bake 1 loaf of bread and one hour to make 1 pie. Chandler takes four hours to bake 1 loaf of bread and four hours to make 1 pie.

- (a) What is Chandler's opportunity cost of baking 1 loaf of bread? What is Monica's?
  - (b) If Monica and Chandler decide to specialize in order to maximize their combined output, who should produce what?
  - (c) Assume that Monica and Chandler each have 10 labor hours. Draw Monica's PPF and make note of her production point with trade.
  - (d) Identify a price at which trade could occur. Assuming that Monica would like to consume 5 pies, use your trade price to find Monica's new consumption point (and label it on your PPF in part c).
4. Some environmentalists want to eliminate all pollution. Use your knowledge of economic decision making to respond to this statement. Be sure to make note of any assumptions you make in formulating your response.